

ADARSH LAL ANILAL, CSSGB

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PROFESSIONAL SUMMARY

Industrial, Manufacturing, Quality, and Process Engineer with an MS in Industrial Engineering and a Certified Six Sigma Green Belt, bringing hands-on experience across manufacturing systems, production planning, process optimization, and quality engineering in high-volume automotive and industrial environments. Skilled in Lean Manufacturing, Six Sigma DMAIC, time studies, line balancing, capacity analysis, root cause analysis, FMEA, SPC, CAPA, and continuous improvement to enhance safety, quality, delivery, throughput, and cost. Experienced in developing standard work, manufacturing documentation, SOPs, BOMs, and routings, designing tooling and fixtures, and analyzing KPIs including OEE, defect rate, and cycle time. Currently building multi-agent AI systems for manufacturing operations using agentic AI, LLMs, and hallucination mitigation techniques. Proven ability to drive efficiency improvements of up to 34% and defect reductions of 90% through data-driven decision-making and cross-functional collaboration.

EDUCATION

Master of Science in Industrial Engineering

Aug 2024 - May 2026

University of Texas at Arlington (UTA), Arlington, TX

- Coursework: Operations Research, Engineering Economy, Statistics & Probability, Linear Regression, Production & Inventory Control, Global Supply Chain Management, Quality Systems, Systems Engineering, Manufacturing Processes, AI in Industrial Applications

Bachelor of Technology in Automobile Engineering

Aug 2019 - June 2023

Manipal Institute of Technology (MIT), Manipal, India

- Coursework: Manufacturing Engineering Technology, Automotive Engines, Ergonomics in Automotive Design, Finite Element Methods, Automotive Transmission Systems

PROFESSIONAL EXPERIENCE

Research Assistant

Aug 2026 - Present

University of Texas at Arlington, Arlington, TX

- Architecting and developing ShiftOps AI, a multi-agent manufacturing shift management system using agentic AI with five specialized agents (extraction, clarification, hallucination detection, verification, escalation) powered by LLMs (Grok/Qwen, OpenAI/GPT)
- Designed a hallucination mitigation pipeline using an independent judge model with verbatim quote verification; unsupported claims are automatically flagged, corrected, and routed to human review before any action is taken
- Integrated real-time CMMS history, equipment registry, sensor data, spare parts inventory, and maintenance records for cross-verification of shift reports across a 16-machine simulated manufacturing plant
- Built interactive CLI pipeline using Rich and Streamlit dashboard for shift report processing, sandbox work order generation, technician assignment, and full audit trail visualization with voice input capability

Research Assistant

Aug 2025 - May 2026

University of Texas at Arlington, Arlington, TX

- Developed agent-based simulation models using NetLogo and HubNet to evaluate process variation, capacity, and operational efficiency under dynamic demand constraints
- Collected, analyzed, and reported production and performance data using KPI analysis, capacity studies, and statistical methods to support continuous improvement and cost reduction
- Collaborated with cross-functional teams to identify root causes, develop loss profiles, and deliver data-driven recommendations to project leadership

Service Advisor Intern

June 2023 - Jan 2024

Jawaher Gulf Garage W.L.L, Sitra, Bahrain

- Led customer complaint root cause and corrective action investigations through on-the-floor process reviews, reducing diagnostic time and improving service quality
- Identified process improvement opportunities and built stakeholder partnerships across operations, supporting revenue growth and operational consistency

Project Intern

Jan 2023 - Apr 2023

JCB India Limited, Jaipur, Rajasthan

- Increased production output by 34% through systematic bottleneck analysis, time-and-motion studies, line balancing, and assembly-line rebalancing for Backhoe Loader production
- Created and maintained work standards, SOPs, BOMs, and routings; designed plant and assembly-line layouts, material flow plans, tooling, and fixtures for a 12,000-unit annual capacity
- Conducted capacity analysis, work sampling, and process flow auditing to enhance quality, throughput, and cost reduction using Lean Manufacturing and continuous improvement tools
- Reduced supply chain costs through strategic Make vs. Buy analysis on Bill of Materials (BOM) components and Value Stream Mapping (VSM), shortening lead times for new production lines
- Collaborated cross-functionally with design engineering and suppliers, ensuring compliance with safety and quality standards and supporting smooth transitions from design to production

Intern

May 2022 - June 2022

Nissan Motors, Sitra, Bahrain

- Applied Root Cause Analysis (RCA), FMEA, and Corrective & Preventive Action (CAPA) to diagnose and resolve complex quality and equipment failures
- Performed dimensional and functional inspection, troubleshooting, and diagnostics, contributing to technical documentation and continuous improvement initiatives

PROJECTS

ShiftOps AI — Multi-Agent Manufacturing Shift Management System

- Built a production-grade multi-agent system where a supervisor's informal shift report flows through Extraction → Clarification → Hallucination Check → Verification → Escalation, with each agent as an independent LLM-powered module
- Implemented tool-use architecture integrating equipment registry, CMMS, sensor trends, spare parts, and technician scheduling; sandbox work orders and notifications ensure no real-world side effects during testing

ShiftReport AI — Intelligent Shift-Handover Report Generator

- Developed an AI-powered web application using GPT-4 and Streamlit that converts unstructured supervisor notes into standardized shift-handover reports with machine IDs, severity classification, and pending action items for manufacturing plants
- Engineered iterative prompt designs and implemented hallucination mitigation by restricting model output to a predefined equipment list, then validated quality with human-in-the-loop rubric scoring and LLM-as-a-Judge detection

Defect Reduction using Lean Six Sigma DMAIC

- Led a Lean Six Sigma DMAIC project reducing defect rate from 10% to <1% across 12 pilot lots, achieving a 90% improvement in product quality within a Quality Management System framework
- Analyzed 12 months of production data using p-charts, Pareto analysis, process mapping, and root cause analysis; sustained sub-1% performance using SPC, TQM, and process capability evaluation

Systems Requirements Modeling (SSM & SysML)

- Improved system clarity and requirements traceability per INCOSE/ISO 15288 standards using SysML artifacts and Soft Systems Methodology (SSM), establishing full traceability across requirements, use cases, and subsystems

Inventory Optimization for Retail Operations

- Engineered and evaluated EOQ-based vending and Continuous Review (Q,r) kiosk systems, recommending an automated tracking solution to reduce stockouts and increase revenue through demand forecasting

Predictive Cost Modeling & Regression Analysis

- Built optimized Simple and Multiple Linear Regression models using SAS for cost estimation, validated with R-squared, ANOVA, residual analysis, and multicollinearity checks

Engineering Economy & Capital Investment Analysis

- Constructed a 15-year cash-flow model using NPV, IRR, and Benefit-Cost Ratio (BCR) analysis, executing sensitivity and scenario analyses to support capital investment and cost-reduction decisions

Aircraft Price Prediction using Regression Modeling

- Built Simple and Multiple Linear Regression models using SAS to predict aircraft pricing, validated with R-squared, ANOVA, residual analysis, and multicollinearity checks; applied hypothesis testing to derive actionable valuation insights

PUBLICATIONS

Shaikh, Md P.; **Anilal, A. L.**; Kulkarni, P.; Krejci, C. Teaching Supply Chain Collaboration Concepts Using Networked Participatory Agent-Based Modeling. *IISE Annual Conference Proceedings*, 2026, 1–6. https://doi.org/10.21872/2026IISE_16525

Anilal, A. L.; Leboulluec, A. K. Comparative Evaluation of Machine Learning Regression Techniques for Predicting CO₂ Emissions in Light-Duty Vehicles. *Int. J. Comput. Appl.* 2026, 187 (114). <https://doi.org/10.5120/ijcaccdf069ed3ae>

TECHNICAL SKILLS

Methodologies: Lean Manufacturing, Lean Six Sigma (DMAIC), Kaizen, Continuous Improvement, Value Stream Mapping (VSM), 5S, Time Studies, Time-and-Motion Studies, Work Sampling, Line Balancing, Capacity Analysis, Root Cause Analysis (RCA), FMEA/DFMEA/PFMEA, Statistical Process Control (SPC), Design for Manufacturing (DFM/DFA), CAPA, Total Quality Management (TQM), Process Flow Auditing, JIT, Kanban, SOP Optimization

Software & Tools: Python, SAS, MATLAB, SAP, Power BI, Minitab, NetLogo (Simulation), SolidWorks (CAD/3D Modeling), AutoCAD, SysML, Streamlit, Groq API, OpenAI API (GPT-4), Rich (CLI), Microsoft Office (Excel-Advanced/Pivot Tables, Access, Word, PowerPoint, Project), ERP Systems

AI & Data: Multi-Agent Systems, Agentic AI Architecture, LLM Integration (Groq/Qwen, OpenAI/GPT), Prompt Engineering, Hallucination Mitigation, LLM-as-a-Judge Evaluation, Tool-Use Orchestration, CMMS Integration

Standards & Compliance: ISO 9001, Quality Management Systems (QMS), INCOSE/ISO 15288, Standard Operating Procedures (SOPs), Standard Work, Bill of Materials (BOM), Routings, Technical Documentation, Safety Compliance

Metrics: Cycle Time, Overall Equipment Effectiveness (OEE), Defect Rate, First Pass Yield (FPY), Throughput, Labor Efficiency, Cost Savings, ROI, NPV/IRR/BCR

Core Competencies: Manufacturing Systems & Process Optimization, Production Planning, Quality Assurance & Control, Process Improvement, Cost Reduction, Supply Chain Management, Plant & Assembly Line Layout, Material Flow Analysis, KPI Analysis, Production Data Analysis, Manufacturing Troubleshooting, Cross-Functional Collaboration, Project Management, Data-Driven Decision Making

AWARDS

- Don Liles Endowed Scholarship (UTA) — March 2025
- Founder's Arena Arlington FIFA AI Hackathon — Second Prize, June 2026

CERTIFICATIONS

Six Sigma Green Belt (CSSGB) • McKinsey Forward Program